

Coding & Solution

Date	29/09/2026
Team ID	UNM1425comiccraft
Project Name	comiccraft
Maximum Marks	5 Marks

Coding & Solution

This section evaluates the quality and completeness of your implemented code and the overall solution. Your submission should demonstrate working functionality, clean code practices, and alignment with your defined requirements.

Instructions:

1. Ensure your code is well-structured, commented, and follows best practices for your chosen technology stack.
2. The solution should address the core problem statement and implement the key functional requirements.
3. Include all source files, configuration files, and setup instructions needed to run the application.
4. Attach or link to your code repository (e.g., GitHub) in the table below.

Solution Summary

Field	Details
Repository Link / URL	<a href="https://github.com/<your-username>/comiccraft">https://github.com/<your-username>/comiccraft
Programming Language(s)	Python, HTML, CSS (Jinja2 templates)
Framework(s) Used	FastAPI, Uvicorn, Jinja2 Google Gemini API (Gemini 1.5 Flash and 1.5 Pro) Hugging Face Diffusers (Stable Diffusion v1.5) with PyTorch FPDF
Key Features Implemented	• Input form: story prompt, character name, setting, tone and art style • 5-panel comic outline generation using Gemini Flash • Narration and character dialogue generation using Gemini Pro • Comic-style panel illustrations using Stable Diffusion • Layout builder and panel-by-panel comic preview page • PDF export using FPDF, followed by an export success page • API routes: /generate, /generate-comic/json, /test-image
Pending / Incomplete Features	• User accounts and a saved comic library • Multi-page story arcs • Cloud deployment (currently runs locally only)
Setup / Run Instructions	1. Create and activate a virtual environment (python -m venv env) 2. Install dependencies: pip install -r requirements.txt 3. Add GEMINI_API_KEY and HF_API_KEY to a .env file 4. Start the server: uvicorn app.main:app --reload 5. Open http://127.0.0.1:8000 (API docs at /docs)

Code Quality Checklist

S.No	Criteria	Status (Yes / No)
1	Code is modular and organized into functions / classes	Yes
2	Meaningful variable and function names are used	Yes
3	Code includes comments / documentation where necessary	Yes
4	Error handling is implemented for critical operations	Yes
5	The application runs without critical errors	Yes
6	Code is committed to a version control repository	Yes

Additional Notes / Comments:

ComicCraft is a FastAPI web application that turns a user's story prompt into a 5-panel AI comic: Gemini Flash creates the outline, Gemini Pro writes the narration and dialogue, and Stable Diffusion draws each panel. The comic can be previewed in the browser and downloaded as a PDF. The application was tested locally with Uvicorn; API keys are read from environment variables (.env).